The Problem Space & Foundational Purpose

Overcoming the failure points of traditional nutrition tracking through automated intelligence

⚠ Traditional Nutrition Tracking Failures

Critical friction points causing user abandonment

- **70%+ Two-Week Abandonment:** Manual logging requires weighing ingredients and searching sprawling databases, creating unbearable user friction.
- **30–50% Estimation Drift:** Guessing portion sizes, cooking oils, and mixed meals introduces severe caloric reporting discrepancies.
- **Static & Impersonal Regimens:** Generic diet templates fail to synchronize with personal Basal Metabolic Rate (BMR) and dynamic lifestyles.

Industry Reality: `High Attrition Rate` `Manual Fatigue` `Uncalibrated Plans`

🎯 The Paradigm Shift: Manual vs. DietAI

Transforming a 15-minute chore into an instant visual capture

TRADITIONAL MANUAL FLOW

1. Measure meal on kitchen scale
2. Search text database (10k+ items)
3. Manual gram input ➔ High Dropout ❌

DIET-AI SMART PIPELINE

- 📷 1-Tap smartphone camera scan
- 🧠 Depth-aware portion segmentation
- ⚡ Instant Macros Logged in < 2s ✓

🎯 CORE PURPOSE OF THE PROJECT

"To eliminate the friction of personal nutrition tracking by uniting computer-vision meal analysis, Mifflin-St Jeor metabolic planning, and safety-guided artificial intelligence into an effortless mobile companion."

0:00 - 0:35 SPEAKER SCRIPT

*"Distinguished jury members, over **70% of individuals abandon calorie tracking within two weeks**. The cause is not lack of commitment, but friction—weighing food, searching databases, and misestimating hidden calories. **The core purpose of DietAI** is to eliminate this barrier completely. By combining real-time computer vision with dynamic, metabolic-aware AI planning, our mission is to make scientific nutrition tracking effortless, accurate, and sustainable for everyone."*

Technical Objectives & AI Innovation Pillars

Engineering three interconnected architectural layers to achieve reliable, safe dietary guidance

1. Real-Time Computer Vision & Depth Analysis

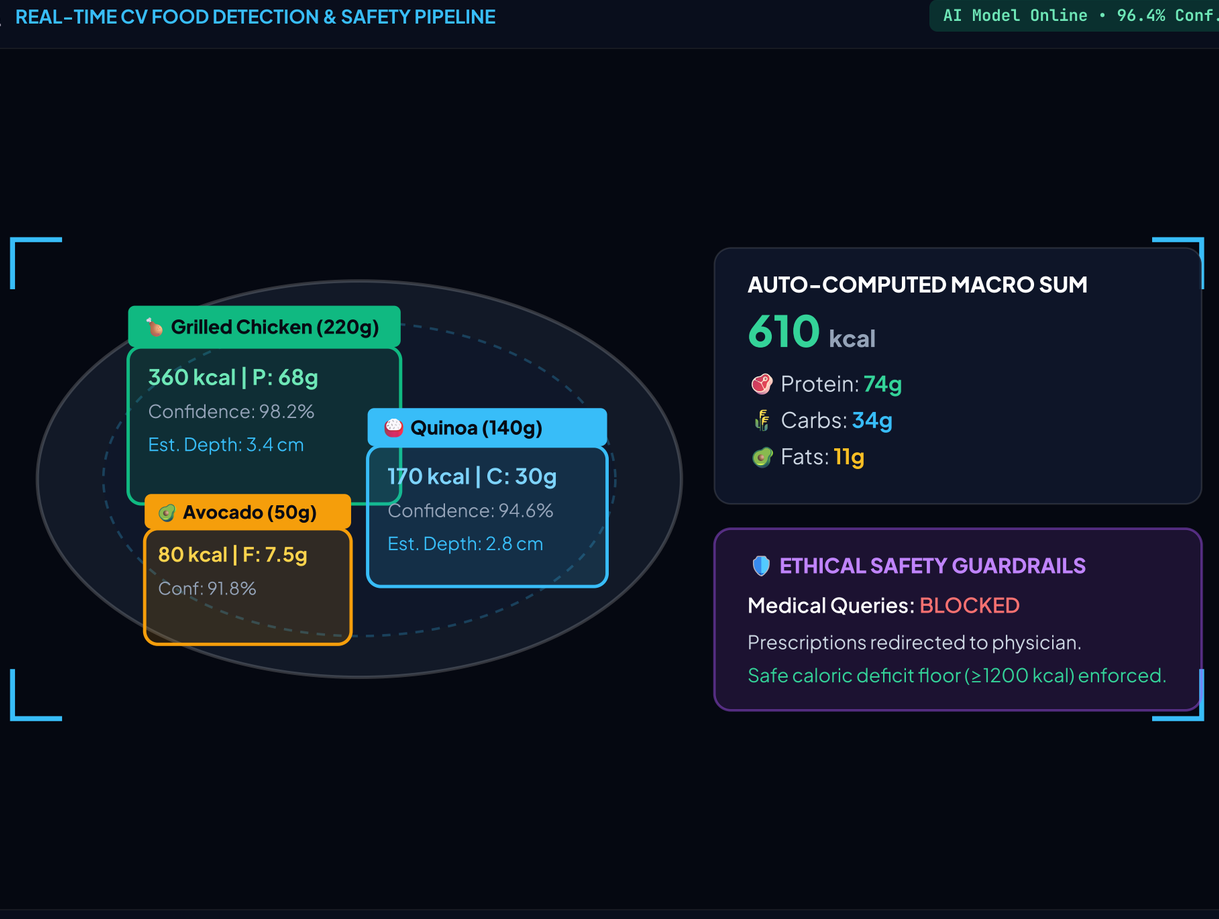
Detects multiple food classes in a single camera frame. Leverages **contour volumetric heuristics** to estimate portion weight without requiring physical kitchen scales.

2. Algorithmic BMR & Dynamic Planning

Calculates exact Basal Metabolic Rate via the **Mifflin-St Jeor formula**. Dynamically generates allergen-safe, goal-oriented (loss/gain/maintenance) weekly meal plans.

3. Ethical Conversational AI & Guardrails

Multi-turn dietary advisor powered by **Gemini API**, reinforced with **strict safety guardrails**: strictly rejects medical diagnoses and prohibits dangerous caloric deficits.



0:35 - 1:15 SPEAKER SCRIPT

"To achieve this purpose, our engineering design focuses on three innovative technical objectives. First, our **computer vision module** *uses the smartphone camera to recognize food items and utilizes depth-based estimation to approximate portion size, translating visual data directly into macronutrients. Second, our* **personalization engine** *calculates exact Basal Metabolic Rate using the Mifflin-St Jeor equation to generate dynamic, allergy-safe weekly meal schedules. Third, our* **conversational AI leverages Gemini with strict safety guardrails** *—actively rejecting medical diagnostic queries to guarantee zero clinical liability."*

Target Deliverables & Real-World Impact

Delivering a verifiable, production-grade mobile prototype with measurable health benefits

Full Mobile Application Prototype

Cross-platform iOS & Android via React Native & Expo

- **Smart Camera Scanner:** Instant capture with OpenCV & AI analysis sheet.
- **Dynamic Calorie Dashboard:** Real-time circular progress & macro split bars.
- **Safety-Guarded Chatbot:** Context-aware dietitian powered by Gemini.

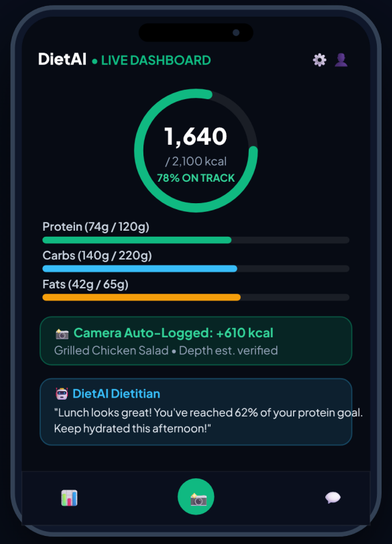
Target Beneficiaries & Value

Empowering sustainable, everyday health decisions

Weight Managers
Effortless caloric deficit, surplus, or maintenance control.

Allergy Conscious
Automated weekly meal plans filtering out allergens.

< 2s	100%	BMR
CV SCAN SPEED	MEDICAL SAFETY	DYNAMIC PLANS



Frontend (Bertan Beyal)
React Native, Expo, TypeScript with live camera viewfinder, daily macro rings, and interactive bottom sheets.

Backend & DB (Eren Torman)
FastAPI RESTful API gateway, PostgreSQL schema, JWT authentication, and automated BMR calculation.

AI & CV Service (Ali Eren Özer)
OpenCV food detection, depth portion heuristics, and Gemini API with zero-medical-diagnosis guardrails.

1:15 - 2:00
SPEAKER SCRIPT

*"Our tangible project outcome is a **production-ready mobile application prototype** uniting our React Native frontend, FastAPI backend, and Gemini AI pipeline. The primary beneficiaries are individuals seeking sustainable lifestyle habits without the exhaustion of manual logging. By compressing a 15-minute daily chore into a **sub-second camera capture** while upholding strict ethical safety boundaries, DietAI delivers an academically robust, technically innovative, and user-centric software solution. Thank you for your time, and we look forward to your questions."*